Computable Ground-Energy Brackets with Finite Null Constraints

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Abstract

Under the stated complete theta–Gamma–prime form realization, the centered ground energy is a computable real: for each positive rational accuracy there is a terminating procedure returning a rational interval of that width containing it. The construction combines an effectively enumerable centered form core, a semidecision for strict negative trials, and finite positive-reference matrix certificates for uniform shifted coercivity. Finite exact derivative-radical constraints are permitted but are not assumed complete. Two overlapping thresholds prevent equality at a tested threshold from blocking the enclosure. The result is conditional on the explicitly identified realization and computable arithmetic inputs; no implementation, useful complexity bound, actual sign, exact-zero decision, or proof or disproof of the Riemann hypothesis is claimed.

Keywords: computable analysis; theta kernel; quadratic forms; finite null constraints; spectral enclosure.

1 Setting, inputs, and statement

For $x \geq 0$ set $\tau_n(x) = \pi n^2 e^{2x}$ and define the positive even kernel and weight

$$\kappa(x) = e^{x/2} \sum_{n\geq 1} (4\tau_n(x)^2 - 6\tau_n(x)) e^{-\tau_n(x)}, \qquad W = \frac{\kappa}{2\cosh(x/2)}.$$

Let $\mathcal{H} = L^2_{\mathrm{even}}(\mathbb{R}, dx/W)$, $m(u) = \int \cosh(x/2) u(x) dx$, $\mathcal{H}_c = \ker m$, and $Cu = u - 2m(u)\kappa$. The inner product is conjugate-linear in the first argument, ordinary Fourier transform is unitary, and $\|\cdot\|_2$ is ordinary $L^2(dx)$. The complete numerator form is

$$\begin{aligned} q_0[u] &= t_\Gamma[u] - c_\Gamma \|u\|_2^2 - p[u], \quad c_\Gamma = \log\pi + \gamma_E + \pi/2 + 3\log 2, \\ t_\Gamma[u] &= \int_0^\infty \frac{e^{-s/2}}{1-e^{-2s}} \|u(\cdot + s) - u\|_2^2 ds = \int_{\mathbb{R}} d_\Gamma(t) |\widehat{u}(t)|^2 dt, \\ p[u] &= 2\,\mathrm{Re} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} \int u(x + \log n) \overline{u(x)} dx. \end{aligned}$$

All prime powers and both orientations are retained. We assume the actual complete closed-form realization and identities in the accompanying working manuscript [1]: $X = \{u \in \mathcal{H} : t_\Gamma[u] < \infty\}$ is the form domain, q_0 has a lower-bounded self-adjoint representative L on $\mathcal{H}$, $\|\kappa\|_{\mathcal{H}} = 1$, $L\kappa = -\kappa/2$,

and C reduces L to L_c on $\mathcal{H}_c$. The bounded correction in $L = T_\Gamma - D_{\text{full}}$ obeys $\|D_{\text{full}}\| < 10$ in that source. In particular the rational bound 16 used below is safe. These are substantive inputs, not consequences asserted for an arbitrary computable quadratic form. We use the form-defined realization; identification with the closure of an independently defined minimal symmetric restriction is a separate input or theorem and is not asserted here.

The source also proves $r_j = \kappa^{(2j)} - 4^{-j}\kappa \in X_c$, $q_0(r_j, z) = 0$ for all $z \in X$, and linear independence. Thus $r_j \in \ker L_c$, and $\mathcal{N} = \overline{\operatorname{span}\{r_j\}}^{\mathcal{H}} \subseteq \ker L_c$. We do not assume equality. Put $m_0 = \inf \sigma(L_c)$; the bound and a nonzero radical yield $-16 \le m_0 \le 0$.

Theorem 1.1. *Under the stated realization and arithmetic estimates in Section 2, there is a recursive procedure which, for each rational $\varepsilon > 0$, terminates with rationals $L \le m_0 \le U$ and $U - L \le \varepsilon$. For any fixed finite independent bank $r_1, \dots, r_j$ (including the empty bank), the procedure can use exact orthogonality to that bank without assuming it exhausts the kernel.*

The theorem is an existence result for an algorithm, not a claim that a matrix has been evaluated or an implementation run. Its finite certificates and exact arithmetic obligations are stated below. Computability of a real number does not decide whether it is exactly zero.

2 Effective arithmetic and a dense form core

Uniform computability means arbitrarily narrow rational enclosures. The theta series and its fixed derivatives have such enclosures on compact intervals, with effective Gaussian remainders. For $x \ge 0$,

$$W(x) \ge \pi^2 e^{4x} e^{-\pi e^{2x}}, \qquad |\kappa^{(s)}(x)| \le A_s e^{(2s+9/2)x - \pi e^{2x}}, \tag{1}$$

with effectively obtainable rational A_s. To see this, write $\kappa(x) = e^{x/2} \sum_{n \ge 1} P_0(n^2 e^{2x}) e^{-\pi n^2 e^{2x}}$, $P_0(y) = 4\pi^2 y^2 - 6\pi y$, and use

$$P_{s+1}(y) = \tfrac{1}{2} P_s(y) + 2y P_s'(y) - 2\pi y P_s(y).$$

For each fixed s, coefficient bounds and a finite sum plus a geometric Gaussian tail bound give A_s. Reflection handles $x < 0$. The displayed lower bound and derivative envelope make weighted norms, ordinary moments, and compact-frequency Fourier transforms of each fixed r_j computable with explicit integrable tails. On a fixed compact interval, positivity of W and a computable derivative bound allow a positive lower bound to be certified by subdivision, so integrals involving $1/W$ are computable.

The full Gamma symbol has the series and uniform truncation bound

$$d_\Gamma(t) = \sum_{\ell \ge 0} \frac{2t^2}{a_\ell (a_\ell^2 + t^2)}, \quad a_\ell = 2\ell + \tfrac{1}{2}, \quad 0 \le d_\Gamma(t) - \sum_{\ell < L} \frac{2t^2}{a_\ell (a_\ell^2 + t^2)} \le 2T^2 \left(a_L^{-3} + \frac{1}{4a_L^2} \right) \tag{2}$$

for $|t| \le T$. Also $0 \le d_\Gamma(t) \le 17t^2$, $\operatorname{Lip}(d_\Gamma|_{[-T,T]}) \le 34T$, and $d_\Gamma(t) \ge \frac{1}{2}\log(1+4t^2)$. These follow from the series or the difference integral, without truncating the Gamma form; the digamma identity and asymptotics are standard [5]. For integer $N \ge 1$ the complete prime-tail operator has the computable cap [4]

$$\tau_N \le 160(N+1)^2 e^{-\pi(N+1)} \longrightarrow 0. \tag{3}$$

Each finite $\Lambda(n)$ is computable by finite factorization. Choose rational upper endpoints $\bar{\tau}_N \ge \tau_N$ tending to zero.

Lemma 2.1 (Effective ambient and centered cores)**.** *There is a recursive list of even, compactly supported, C^1 piecewise cubic splines with rational knots and coefficients in $\mathbb{Q}+i\mathbb{Q}$, vanishing with their first derivatives at the support endpoints, that is dense in X in the form norm. Its exact images under C are dense in X_c.*

Proof. For $u \in X$, choose even smooth cutoffs χ_R equal to one on $[-R,R]$, supported compactly, and with $\|\chi_R'\|_\infty \le C/R$. Weighted dominated convergence handles $\mathcal{H}$. In the Gamma difference of $(1-\chi_R)u$, the part multiplying $u(x+s)-u(x)$ tends to zero by dominated convergence, while the commutator is bounded in squared norm by

$$\|u\|_2^2 \int_0^\infty \frac{e^{-s/2}}{1-e^{-2s}} \min(Cs/R,2)^2 ds \longrightarrow 0.$$

Thus cutoff is form-convergent. On a fixed compact support, even mollification converges in ordinary Gamma form by the bounded Fourier multiplier and dominated convergence, and in $\mathcal{H}$ because W has a positive minimum on an enlarged compact interval. Hence even compact smooth functions form a core.

A rational grid and rational Hermite endpoint values and derivatives give a recursive list of the displayed splines. Rational approximation to the sampled values and derivatives, plus mesh refinement and even symmetrization, approximates each compact smooth even function in ordinary H^1 on a common compact interval. The compact lower bound on W and $d_\Gamma \le 17t^2$ convert this to form-norm convergence. The exact projection C is form-continuous because m is $\mathcal{H}$-bounded and $\kappa \in X$; applying it to the list gives a centered core. No effective rate of convergence of an arbitrary input to the list is needed. □

Every finite trial value $q_0[v]$, $\|v\|_{\mathcal{H}}$ and $m(v)$ is computable. For a spline v, put $D_v = \|v''\|_1$; two integrations by parts give $|\widehat{v}(t)| \le D_v/(\sqrt{2\pi}t^2)$, and the full Gamma frequency tail is at most $17D_v^2/(\pi T)$. Compact theta integrals and (2) handle retained terms, and (3) pays every omitted prime power. The exact centering identities are

$$q_0[Cv] = q_0[v] + 2|m(v)|^2, \qquad \|Cv\|_{\mathcal{H}}^2 = \|v\|_{\mathcal{H}}^2 - 4|m(v)|^2. \tag{4}$$

The computable real $m(v)$ is not rounded and then treated as an exact zero. For rational $b<0$, enumerate centered trials and precision until the certified upper endpoint of $q_0[Cv]-b\|Cv\|_{\mathcal{H}}^2$ is strictly negative. This is a sound semidecision for $m_0<b$, complete by the variational principle and Lemma 2.1; it need not halt if equality holds.

3 Finite exact constraints and the positive reference

Fix a rational $\sigma>0$ and a finite independent radical bank $r_1,\dots,r_j$, possibly empty. Let P_j be its $\mathcal{H}_c$ projection. As $L_cP_j=0$,

$$L_c+\sigma I = \sigma P_j + (I-P_j)(L_c+\sigma I)(I-P_j). \tag{5}$$

The residual space is $S=\operatorname{span}\{\kappa,r_1,\dots,r_j\}^\perp$ in $\mathcal{H}$; no infinite-radical projection is used.

For integer N with $\bar\tau_N<\sigma$, put $\delta'=\sigma-\bar\tau_N>0$ and define

$$c_N(t) = c_\Gamma + 2\sum_{2\le n\le N} \frac{\Lambda(n)}{\sqrt{n}}\cos(t\log n), \qquad a_N=(d_\Gamma-c_N)_+, \quad b_N=(c_N-d_\Gamma)_+.$$

Let A be the positive ambient representative of

$$p_{\mathrm{ref}}[u] = \int a_N(t)|\widehat{u}(t)|^2 dt + \delta'\|u\|_{\mathcal{H}}^2,$$

and let A_S be its form restriction to S. Since $|a_N - d_\Gamma| \le C_N := |c_\Gamma| + 2\sum_{n\le N}\Lambda(n)/\sqrt{n}$, the domain is exactly X and $A_S \ge \delta' I$. If B_S is the nonnegative form with multiplier b_N on S, the complete tail bound gives

$$q_0[u] + \sigma\|u\|_{\mathcal{H}}^2 \ge p_{\mathrm{ref}}[u] - \int b_N|\widehat{u}|^2 dt \qquad (u \in X \cap S). \tag{6}$$

A strictly supported frequency cutoff is found effectively: choose a rational $\bar{C} > C_N$ and integer T with $\frac{1}{2}\log(1+4T^2) > \bar{C}$. Then $b_N = 0$ for $|t| \ge T$, without deciding any equality at a zero of $d_\Gamma - c_N$. The positive Birman–Schwinger operator $K = A_S^{-1/2} B_S A_S^{-1/2}$ is trace class: its compact-frequency feature representation below has finite integral of squared norms.

Let $V = (\kappa, r_1, \dots, r_j)$, $R = A^{-1}$, and P_S the orthogonal projection onto S. The restricted inverse, extended by zero on $S^\perp$, is

$$R_S = R - RV(V^*RV)^{-1}V^*R = A_S^{-1}P_S. \tag{7}$$

This is the Lagrange-multiplier identity for the restricted form. The finite matrix V^*RV is positive definite because R is positive injective and the columns are independent. The actual weighted Gram, not an identity matrix, is used. Interval refinement eventually certifies a positive lower eigenvalue and its inverse, without an a priori condition number.

4 A finite positive matrix gate

For $t \ge 0$ define $g_t(x) = W(x)\cos(tx)/\sqrt{2\pi}$ and $h_t = A_S^{-1/2}P_S g_t$. For even u, $\widehat{u}(t) = \langle g_t, u\rangle_{\mathcal{H}}$. Thus

$$K = 2\int_0^T b_N(t)\, h_t h_t^* dt. \tag{8}$$

The theta normalization yields $\int W \le 1/4$ and $\int x^2 W \le 1/2$: the latter follows from $2y^2 \le \cosh^2 y$, obtained from the first three even terms of $\cosh^2 y$. Hence with $G_0 = \sqrt{1/(8\pi)}$, $G_1 = \sqrt{1/(4\pi)}$,

$$\|h_t\| \le G_0/\sqrt{\delta'}, \qquad \|h_t - h_s\| \le G_1|t-s|/\sqrt{\delta'}.$$

Let $L_N = 2\sum_{2\le n\le N}\Lambda(n)n^{-1/2}\log n$ and $L_b = 34T + L_N$. Since $0 \le b_N \le C_N$ and $\operatorname{Lip} b_N \le L_b$, r midpoint cells of width $\Delta = T/r$ give

$$K_r = \sum_{i=0}^{r-1} w_i h_{t_i} h_{t_i}^*, \quad w_i = 2b_N(t_i)\Delta, \quad \|K - K_r\| \le \eta_{\mathrm{quad}} := \frac{T\Delta}{2\delta'}(L_b G_0^2 + 2C_N G_0 G_1), \tag{9}$$

$$M_r = \operatorname{diag}(\sqrt{w_i})\,[\langle g_{t_i}, R_S g_{t_\ell}\rangle]_{i,\ell}\,\operatorname{diag}(\sqrt{w_i}). \tag{10}$$

The nonzero spectra of K_r and M_r agree. Square roots of the nonnegative computable weights are computable even when a weight is zero; no zero-mass decision or division is needed. Rational upper enclosures for C_N, L_b, G_0, G_1 are used in η_{quad}.

If $\widetilde{M}$ is a symmetric rational approximation with certified row-sum norm error η_{entries} and

$$I - \widetilde{M} - (\eta_{\mathrm{entries}} + \eta_{\mathrm{quad}})I > 0, \tag{11}$$

then $\|K\| < 1$. The finite inequality can be verified by exact rational LDL or Sylvester tests. It supplies a rational margin $\gamma > 0$, hence (6) gives $q_0 + \sigma\|\cdot\|_{\mathcal{H}}^2 \ge \gamma\delta'\|\cdot\|_{\mathcal{H}}^2$ on S; (5) restores the exact radical bank with minimum margin $\min\{\gamma\delta', \sigma\}$. The gate pays all matrix-entry, scalar, quadrature, and complete prime-tail errors. It is a sufficient finite certificate, not an evaluated one.

5 Computing the constrained inverse entries

The following construction is included so that (10) does not assume an inverse oracle. The previously cited working file [3] exists in the supplied source package; the account here uses the general bound $\bar{W} \geq \|W\|_\infty$ (one may take rational $\bar{W} = 2$), rather than assuming a particular sharper cap.

On the ambient $\mathcal{H}$ let $J : \mathcal{H} \to L^2_{\mathrm{even}}(dx)$ be inclusion, so $\|J\|^2 \leq \bar{W}$, and for $k \geq 1$ put

$$m_k(t) = \min(a_N(t), k)\mathbf{1}_{[-k,k]}(t), \quad S_k = J^*\mathcal{F}^{-1}m_k\mathcal{F}J, \quad A_k = \delta' I + S_k.$$

The positive forms increase to A, so $A_k^{-1} \downarrow A^{-1}$ strongly by monotone convergence of closed forms [6]. With $\alpha_k = \delta' + k\bar{W}$, $Y_k = I - A_k/\alpha_k$, and $q_k = 1 - \delta'/\alpha_k$,

$$P_{k,\ell} := \alpha_k^{-1} \sum_{s=0}^{\ell} Y_k^s, \qquad A^{-1} \leq A_k^{-1} \leq P_{k,\ell} + q_k^{\ell+1}\delta'^{-1} I. \tag{12}$$

Thus the right side gives an upper enclosure for $\langle f, A^{-1}f\rangle$; $P_{k,\ell}$ alone is not claimed to be a lower bound for the full inverse. Expanding the finite polynomial uses the degree-zero term $\|f\|^2_{\mathcal{H}}$ and, for $n \geq 1$, integrals over $[-k,k]^n$:

$$\langle f, S_k^n f\rangle_{\mathcal{H}} = \int_{[-k,k]^n} \overline{\widehat{f}(t_1)}\widehat{f}(t_n) \prod_{h=1}^{n} m_k(t_h) \prod_{h=1}^{n-1} \frac{\widehat{W}(t_h - t_{h+1})}{\sqrt{2\pi}}\, dt_1 \cdots dt_n. \tag{13}$$

The effective theta tails, derivative bounds, (2), and elementary moment bounds give computable moduli for $\widehat{W}$ and $\widehat{f}$. The function $\min(a_N, k)$ is computably Lipschitz on the integration cube $[-k,k]$; the sharp cutoff in m_k can jump at its endpoints, so no global Lipschitz assertion is used. Certified rational integration on these cubes gives arbitrarily tight enclosures.

Independent lower enclosures are obtained from the ambient core in Lemma 2.1:

$$2\,\mathrm{Re}\langle f, v\rangle - p_{\mathrm{ref}}[v] \leq \langle f, A^{-1}f\rangle, \qquad \sup_{v \text{ in core}} \left(2\,\mathrm{Re}\langle f, v\rangle - p_{\mathrm{ref}}[v]\right) = \langle f, A^{-1}f\rangle. \tag{14}$$

The full multiplier a_N is used. For a spline v, its frequency tail is bounded by $D_v^2[17/T + C_N/(3T^3)]/\pi$. Dovetail k, ℓ, trials, and integral precision, retaining the best rational upper and lower endpoints. Strong monotone convergence and core density force the gap below every requested tolerance; the stopping test itself uses rational endpoints only. Polarization computes all entries for the finite family $g_{t_i}, \kappa, r_1, \ldots, r_j$; their transforms and norms are computable by Section 2. Refine V^*RV until its positive definiteness is certified, then apply (7). This proves the needed finite entry procedures terminate; it supplies no useful rate.

6 Strict-shift completeness and equality-safe bracketing

Suppose $q_0[u] + \sigma\|u\|^2_{\mathcal{H}} \geq c\|u\|^2_{\mathcal{H}}$ on X_c for some unknown $c > 0$. For sufficiently large N, $2\bar{\tau}_N < c$, and the finite-head reference difference in (6) has a positive $\mathcal{H}$ lower bound on S. Since K is compact and positive, $\|K\| \geq 1$ would yield an eigenvector contradicting this strict bound. Thus $\|K\| < 1$, and increasing midpoint resolution and inverse-entry precision eventually verify (11). Fair dovetailing over N, all finite matrix refinements, and the terminating inverse-entry jobs gives a semidecision complete for uniform shifted coercivity, equivalently $m_0 + \sigma > 0$. Pointwise positivity without a uniform $\mathcal{H}$ margin is not sufficient for this completeness assertion. Conversely every successful gate is sound by (6). It need not stop when the shifted ground energy is exactly zero.

To prove Theorem 1.1, start from $[L, U] = [-16, 0]$. While $U - L > \varepsilon$, put $a = L + (U - L)/3$ and $b = L + 2(U - L)/3$; then $a < b < 0$. Dovetail the positive gate for shift $\sigma = -a$ and the strict trial search for $m_0 < b$. If the gate succeeds set $L = a$; if a strict trial succeeds set $U = b$. If $m_0 \geq b$, the shifted form has strict margin $m_0 - a > 0$; if $m_0 \leq a$, the strict trial inequality holds for some core vector; in the overlap both branches succeed eventually. In particular equality at a or b never blocks both branches. Each step reduces the interval width by 2/3, so at most $\lceil \log(16/\varepsilon)/\log(3/2) \rceil$ steps are needed when $\varepsilon < 16$. The bound counts outer steps only, not the duration of their inner searches.

A positive finite certificate consists of a rational prime-tail cap, theta/Gamma truncation and integral enclosures, a certified finite constraint inverse Gram, the rational matrix and entry/quadrature error debits, and an exact positive-definiteness witness for (11). A negative certificate consists of a rational spline, (4), and a strict rational upper enclosure for its complete form. A checker verifies finite evidence; no computed projection onto the entire kernel or equality test is used.

7 Boundary of the result and sources

The working manuscript [1] is the input for the form-defined self-adjoint realization, normalization, full-domain radical identity, and complete correction bound (Lemmas a:lem:theta, a:lem:prime, a:lem:radical and the weighted realization). The two supplied notes [2, 3] record the immediate construction; the latter exists as a separate source file and its inverse enclosure mechanism is also given in Section 5. The tail estimate is recorded in [4]. These working documents are not presented as independently published references. The supplied checks test finite algebra and source hashes, not the asserted total-recursive theorem.

No exact test for $m_0 = 0$ follows. If $m_0 < 0$, the strict trial search eventually finds a negative witness; if $m_0 = 0$, finite absolute brackets alone never certify equality. No useful a priori running-time bound, effective infinite-radical repair rate, sign of the actual relative low-collar form, or Riemann hypothesis conclusion is claimed. No historical priority claim is made.

References

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